JIAWEN ZHANG

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EDUCATION

University of Illinois Urbana-Champaign Mar. 2026 – Present
Visiting Researcher, Siebel School of Computing and Data Science *Champaign, IL, USA*

- **Advisors:** Prof. James M. Rehg, Prof. Ismini Lourentzou

Peking University Sep. 2024 – Present
M.S. in Electronic Information, National Biomedical Imaging Center *Beijing, China*

- **Advisor:** Prof. Shuai Na
- **Core Courses:** Large Language Models and Applied Practices, Machine Learning, Natural Language Processing, AI in Medicine, Algorithm Analysis and Design, Software Architecture and Design.

Lanzhou University Sep. 2019 – Jun. 2024
B.Eng. in Computer Science and Technology, UAIS Lab *Lanzhou, China*

- **Advisor:** Prof. Xiaowei Li
- **Rank: 1/168 & National Scholarship** (Average Score: 93.36/100)
- **Core Courses:** Introduction to Artificial Intelligence, Machine Learning, Data Structures, Principles of Computer Organization, Computer Networks, Assembly Language.

RESEARCH INTERESTS

Multimodal AI & World Models for human behavior understanding;
Agentic RL and Post-Training of vision–language models;
Computational Imaging and physics-constrained inverse problems.

PUBLICATIONS

*Equal contribution. †Corresponding author.

- **Jiawen Zhang**, et al. Gaze World Model. Under review at *IEEE/CVF Conference on Computer Vision and Pattern Recognition* **(CVPR)**, 2027.
- **Jiawen Zhang***, Yifan Shen*, Jian Xu, Junho Kim, Ismini Lourentzou, Xu Cao, Meihuan Huang. Evaluating Cognitive Age Alignment in Interactive AI Agents. Under review at *Conference of the European Chapter of the Association for Computational Linguistics* **(EACL)**, 2027; *arXiv preprint* **arXiv:2605.17894**, 2026.
- **Jiawen Zhang**, Han Yang, Jintao Ma, Bangxu Fan, Tong Shi, Xiaoyan Zheng, Songqing Xie, Zhuojun Xie, Shuai Na†. Radon–Full-Waveform Inversion for Suppressing Scalp Reverberation and Skull-Induced Aberration in Transcranial Photoacoustic Computed Tomography. ***Photoacoustics***, 2026.
- **Jiawen Zhang**, et al. Dual-Modality US–PA Aberration Correction for Transcranial Imaging. Under review at *IEEE Transactions on Medical Imaging* **(TMI)**, 2026.
- Yifan Shen, Zhuoqing Zhong, Xinzhuo Li, Jiateng Liu, Bingxuan Li, Wei Cao, Pei Tian, Lin Zhu, Jian Xu, Zewei Chen, Tianjiao Yu, Haichao Zhang, **Jiawen Zhang**, Yuner Zhang, Brian Nlong Zhao, Sen Fang, Nan Huang, Bowen Fang, Chen Fang, Boyi Li, Onkar Susladkar, Frank Yang, Yuanzhe Liu, Jingyuan Zhu, Cheng Qian, Yijiang Li, Xinyang Han, Yuan Shen, Xiang Li, Bolin Lai, Xu Cao, Chengde Wan, Yiwen Song, Yue Guo, Heng Ji, Xuan Di, Yiwei Wang, Yaoyao Liu, Dimitris N. Metaxas, Yun Fu, James M. Rehg, Chengxiang Zhai, Ismini Lourentzou. Native Real-Time Interactive Omni Intelligence: A Unified Framework. *Survey, Zenodo preprint* **doi:10.5281/zenodo.23404586**, 2026.

RESEARCH EXPERIENCE

Rehg Lab, University of Illinois Urbana-Champaign Advisors: **James M. Rehg, Ismini Lourentzou** Mar. 2026 – Present
Research Intern Mentors: **Xu Cao**, **Houze Yang** *Champaign, IL, USA*

Gaze World Model (GWM) — Project Lead *Jul. 2026 – Sep. 2026*

- **Problem:** Formulated person-specific gaze modeling in third-person video as a **world prediction problem**, unifying gaze-point estimation, in-clip gaze-shift detection, and forecasting of future gaze targets and shift events.
- **Model:** Developed **GWM**, combining a frozen V-JEPA 2.1 encoder, RoPE-based scene–head cross-attention, a 3D-RoPE temporal trunk, and a latent world predictor trained with a **motion-weighted latent prediction loss** (no pixel reconstruction).
- **Data:** Built a human-in-the-loop annotation engine (rule proposals, multimodal-model verification, human review) that converts frame-level labels in VideoAttentionTarget and ChildPlay into **6,204 event-level gaze shifts** and 15,086 training windows.
- **Results:** **State of the art** over Gaze-LLE, PaGE, SAGE, Qwen3.8 and Gemini 3.8: shift-detection AP **78.38 → 89.72**, future-shift AP **66.23 → 76.18**, future gaze L2 **0.123 → 0.104**; the **105M** ViT-B variant beats the **1.7B** PaGE ViT-H+ on every shift metric.

National Biomedical Imaging Center, Peking University Advisor: **Prof. Shuai Na** Oct. 2024 – Mar. 2026

Graduate Researcher *Beijing, China*

Radon–FWI for Transcranial Photoacoustic Tomography — First Author *Oct. 2024 – Jul. 2025*

- **Problem:** Formulated transcranial PACT as a **PDE-constrained nonlinear inverse problem**, explicitly modeling skull heterogeneity, multipathing, elastic scattering conversion, and scalp-dominated reverberations.
- **Method:** Proposed a dual-domain **RT–FWI** framework: a high-resolution linear Radon transform separates scalp multiples by slowness (no prior SOS needed), and the resulting masks act as a **scalp-energy regularizer** in elastic FWI, reducing non-uniqueness and cycle-skipping.
- **Implementation:** Adaptive Radon-domain masking (block-wise Otsu + morphological cleaning) and **8th-order staggered-grid FDTD** with CPML, optimized via adjoint gradients and FISTA with restart.
- **Results:** Improved axial resolution from **1.14 mm → 0.28 mm** and SSIM from **0.42 → 0.88**.

UMI-PAI: Dual-Modality US–PA Aberration Correction — First Author *Aug. 2025 – Mar. 2026*

- **Method:** Proposed **UMI-PAI**, a measurement-based framework that estimates skull-induced **phase distortions** from the ultrasound reflection matrix and applies the correction operator to photoacoustic reconstruction at minimal computational cost.
- **Optimization:** Developed a spatially adaptive phase-screen model with **iterative reflection-matrix optimization** (adaptive windowing, locally weighted updates), integrated into Fourier-domain PA reconstruction via phase-conjugation compensation.
- **Results:** Validated on acrylic phantoms and murine skulls: up to **44% PSNR gain** (**39.30 dB**) over baselines, **without CT/MR priors** or costly elastic-wave inversion.

INDUSTRY EXPERIENCE

PediaMed AI Mar. 2026 – Jun. 2026

Research Intern Mentor: **Yifan Shen**

ChildAgentEval: Cognitive Age Alignment of AI Agents — First Author *Mar. 2026 – Jun. 2026*

- **Benchmark:** Built **ChildAgentEval**, a WISC-inspired interactive web benchmark that evaluates the **cognitive age alignment** of multimodal LLM agents across psychometrically grounded cognitive domains.
- **Method:** Proposed a **two-stage skill-guided distillation** strategy that turns empirical child developmental markers into executable constraints on agent language, memory capacity, and task-solving behavior.
- **Findings:** Across proprietary and open-weight models, cognitive filters yield **monotonic, age-ordered** score and linguistic progressions, while exposing limits in calibrating **working memory** and **visuospatial reasoning**.

Tencent Holdings Limited Dec. 2025 – Mar. 2026

Research Intern, Multimodal Large Models Leaders: **Qi Xiong**, **Yingsong Huang**

- **Distillation:** Distilled **127K** structured CoT trajectories from a **397B** teacher into a **4B multimodal model** via multi-task SFT, fusing text, DOM trees, and visual inputs for generative web security assessment.
- **Agentic RL:** Built a **GRPO** post-training pipeline with **dynamic prompt sampling** (prioritizing medium-difficulty tasks by success-rate variance) and a composite reward on reasoning fidelity, chain density, and length to curb reward hacking.
- **Results:** Cut tool-call rounds **5.43 → 2.90** and end-to-end latency by **70.8%**, while raising decision accuracy to **88.50%**, **surpassing the teacher** model.
- **Robustness:** Designed fallback reasoning for crawler failures and visual-input timeouts, reducing false positives by **65.9%** across **221K** live traffic samples.

HONORS & AWARDS

2025 **Peking University Social Scholarship**, Peking University
2023 **Finalist (F Award)**, Mathematical Contest in Modeling (MCM)
2022 **National Encouragement Scholarship**, Ministry of Education of China
2021 **National Scholarship**, Ministry of Education of China
2021 **Outstanding Student Model**, Lanzhou University

LEADERSHIP

Organization Commissary, School of Software and Microelectronics, Peking University 2024 – Present
Deputy Class Advisor, Computer Science Cohort 2022, Lanzhou University 2022 – 2023
President, Student Association Management Center, Youth League Committee, Lanzhou University 2021 – 2022
Department Officer, Student Association Management Center, Lanzhou University 2020 – 2021

SKILLS

Languages: English (Fluent), Chinese (Native)
Programming: Python, C/C++, CUDA, Verilog, MATLAB
Frameworks: PyTorch